Kevin Ezzedine

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Education

Polytechnique Montréal GPA: 3.89/4.3
B.Eng., Software Engineering *Aug. 2023 – May 2027*

Université de Montréal GPA: 4.3/4.3
Minor in Mathematics *Sept. 2024 – Aug. 2027*

Experience

Axians Canada
Full-Stack Software Developer *May 2025 – December 2026*

- Improve **system scalability** and **team productivity** across every department by designing and implementing a microservices architecture adopted company-wide (**300 people**).
- Ship **full-stack** features by designing the architecture (**NestJS**/**Angular**, **Prisma**, **Azure SQL**) and automating CI/CD with **GitHub Actions**.
- Deploy over **twenty** web and mobile features across **8 applications** by writing SRS and software architecture documents on Azure.
- Expand the portfolio to **9 delivered applications** by collaborating with development teams based in **France** on a joint accounting project.
- Secure the platform by implementing **SSO Entra ID**, AES-256-GCM encryption, and **XSS**/**CSRF** protections.

DAIS — Polytechnique Montréal
Research Intern — Factorized Query Engine FFX (C++20) *May 2026 – October 2026*

- Speed up complex filter processing by up to **50x** by designing a caching system integrated into a *push-based* architecture, enabling reuse of high-cost subtrees.
- Extend the engine's capabilities by implementing **2** new pipelined operators (`InLJoinWithCache` and `Probe`), along with a *chained hashing* table for the caching system.
- Support linear and *branch queries* by extending the engine's control flow, including projection and intersection operations.

Polytechnique Montréal
Lab Instructor — INF3710 (Software Systems Design and Development) *August 2026 – December 2026*

- Improve students' understanding of course concepts by designing and running weekly lab sessions, including presentation slides and interactive **Kahoot** quizzes.
- Reduce assignment grading time (**C++**, **Python**, **SQL**, **DuckDB**) by developing an automated grading pipeline on **Gradescope**, standardizing evaluation.
- Present course content to over **200 students** across **4 groups**, on a weekly basis.

PolyMath — Mathematics Club
Founder & President *May 2026 – Present*

- Create the first mathematics club at Polytechnique Montréal by founding and structuring PolyMath with a founding team of about **ten** members.
- Drive high-level mathematical competition by organizing a tournament (MIT Integration Bee style) on the theme of **sustainable development**.
- Diversify the club's activities by organizing a **discussion panel**, in addition to competitions and research activities.
- Contribute to undergraduate research by completing **2 research projects**, including a study on 3-way comparison sorting and a graph approximation study (feedback vertex sets in bipartite graphs).

PolyAI
Artificial Intelligence Instructor *May 2025 – Present*

- Unblock students' progress on difficult concepts by running targeted comprehension workshops.
- Support students through modern digital tools by designing and delivering about **ten** interactive educational resources.
- Advance **CodeML** competition participants by clarifying the machine learning concepts needed to move their projects forward.

Google Developer Group — Polytechnique Montréal

VP of Training *Dec. 2025 – May 2026*

- Relaunch the **Google Developer Group** chapter, inactive for several months, after being recruited to rebuild the organizing team.
- Draw over **100 people** per workshop from the relaunch onward by designing and running **3 technical workshops** (AI, Cloud, DevOps, Mobile) per session.
- Introduce **RAG** (*Retrieval-Augmented Generation*) systems, a topic still little known at Polytechnique, by leading a workshop with a live implementation demo.

Propolys Medtech Program

Entrepreneur *Dec. 2025 – May 2026*

- Turn a medical concept into a viable business venture by collaborating alongside **18** selected entrepreneurs, building market validation skills.
- Confirm the concept's relevance by consulting **15** healthcare professionals during the validation process.
- Validate the project's viability by presenting it to a panel of **5 people** at the program's conclusion.

Technical Skills

Languages: TypeScript, JavaScript, Python, SQL, Java, C/C++
Frameworks & Platforms: NestJS, Angular, Node.js
AI / ML: scikit-learn, PyTorch, TensorFlow; OpenCV
Data / Libraries: Pandas, NumPy, Matplotlib, Pillow
Cloud & DevOps: Microsoft Azure (App Service, Entra ID SSO, Azure SQL, Blob Storage, VM, Container Apps), GitHub Actions (CI/CD), Docker, Linux, Git
Databases / ORM: Azure SQL (SQL Server), SQLite, DuckDB, Prisma ORM
Security: field-level encryption with AES-256-GCM, JWT
Mathematics: Linear algebra, discrete mathematics, multivariable calculus, graph theory; probability & statistics, analysis I

Projects

Adaptive Distribution Search **(ADS)** | *C++* August 2026 – Present

- Optimize the allocation of a research budget by designing a **stochastic** method for different optimization families (***Derivative-Free Optimization***, **combinatorial optimization**), modeling each move's potential gain as a **random variable**.
- Speed up **convergence** toward an optimal solution by leveraging **probability distributions** to dynamically prioritize the most promising search regions.
- Prepare the submission of a **research paper** to a relevant conference by conducting a **state-of-the-art** (*SOTA*) review and coordinating a team of contributors.

Awards & Honors

UPIR Scholarship — Undergraduate Research Initiation in Mathematics (Polytechnique Montréal)

UPIR Scholarship — Undergraduate Research Initiation in Artificial Intelligence (Polytechnique Montréal)